Lab Animal Imaging

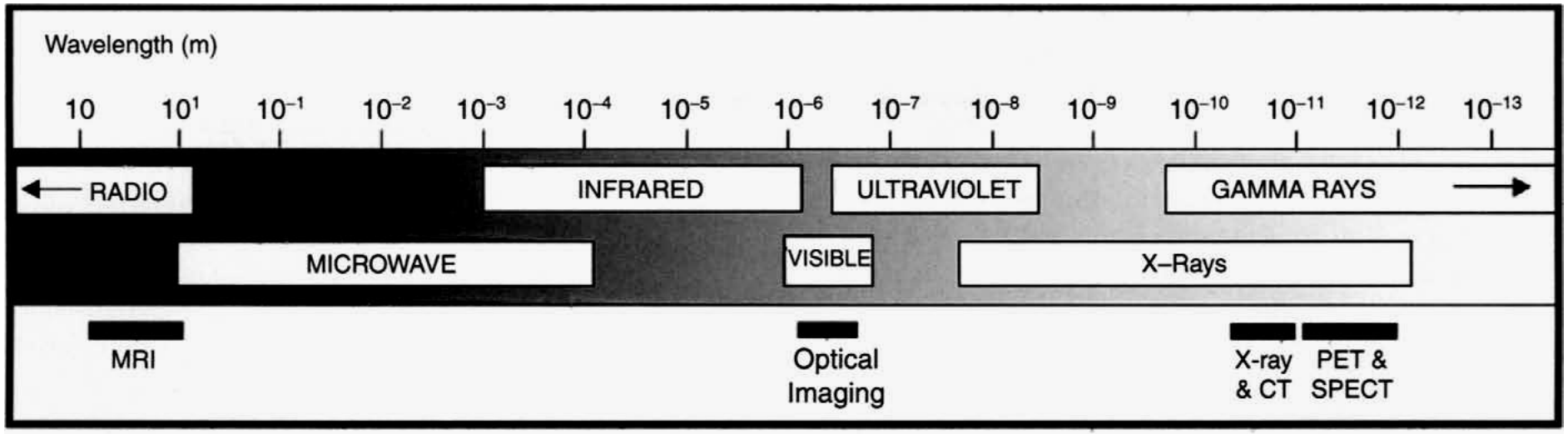
萬灼華
臺大獸醫專業學院

10/22/2026

Imaging techniques

- Experimental & Diagnostic (in lab animal field)
- ✓ Animal size!
- Radiograph (plain & contrast)
- Fluoroscopy
- X-ray Computed Tomography (CT) [Dynamic contrast CT]
- Ultrasonography
- Magnetic Resonance Imaging (MRI)
- Positron Emission Tomography (PET)
- Single photon Emission Computed Tomography (SPECT)

Electromagnetic Spectrum for In-vivo Imaging



MRI: Magnetic resonance imaging
PET: Positron emission tomography
SPECT: Single photon emission computed tomography
CT: X-ray computed tomography

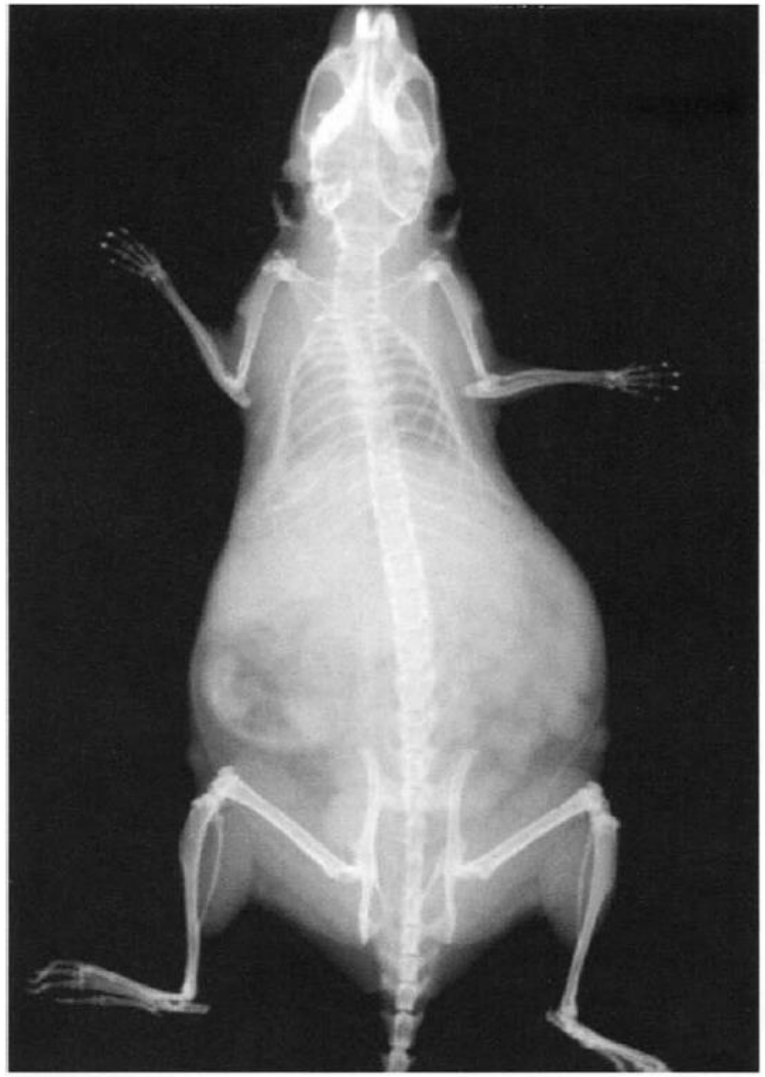
X-Ray Imaging

- To use x-rays relying on the differential absorption of x-rays by tissues of various density & chemical composition
- more x-ray are absorbed by higher density tissues (bone) and fewer by lower density tissues (lung)
- including x-ray computed tomograph

Been applied to improve/expand the usage and to develop new ones

- Radiograph
  - X-rays w/ high-speed film in human dentistry also been used in lab animals
  - Specialized X-ray equip/film/cassettes to get high-definition images of soft tissue (human mammography), also used in mice/rats
  - Fluoroscopy, a modification of X-ray examination, used to visualize the continuous movement of internal structures;

    eg. Barium sulfate-tantalum paste adheres to soft tissues to increased visualization.



A



B

Fig. 14-3 X-ray radiograms of a mouse. (a) Whole-body image, with the display windowed such that both bone and soft tissue can be seen (image courtesy of Stephen Rendig, UC Davis). (b) Image of mouse kidneys following administration of a vascular contrast agent. (Image courtesy of Faxitron X-Ray Corporation, Wheeler, IL.)

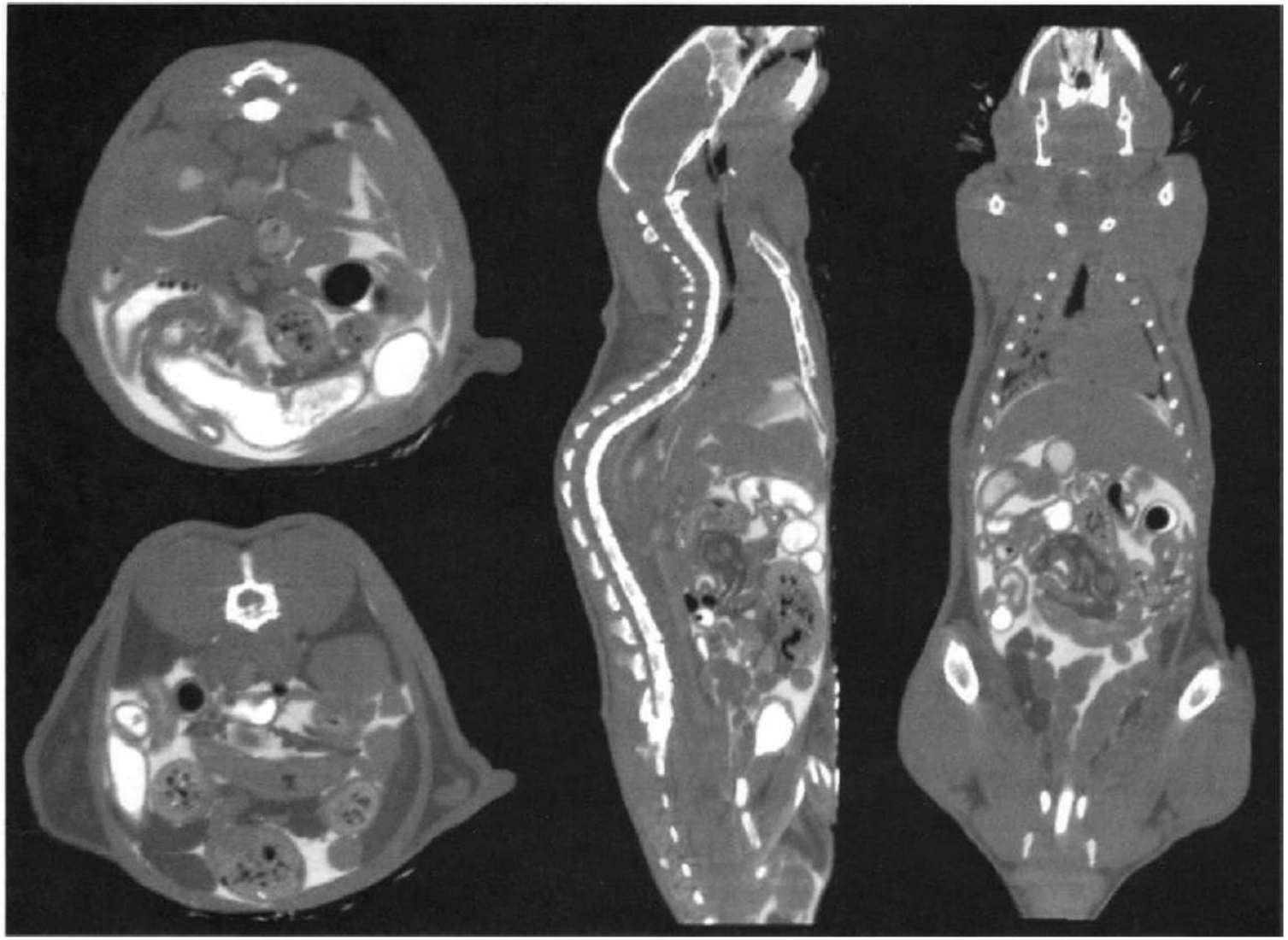


Fig. 14-4 Whole-body CT imaging of a mouse using both oral and IP contrast agents. Transaxial, coronal, and sagittal sections are shown. (Images courtesy of Siemens Preclinical Solutions, Knoxville, TN.)

High-resolution 3D MicroCT

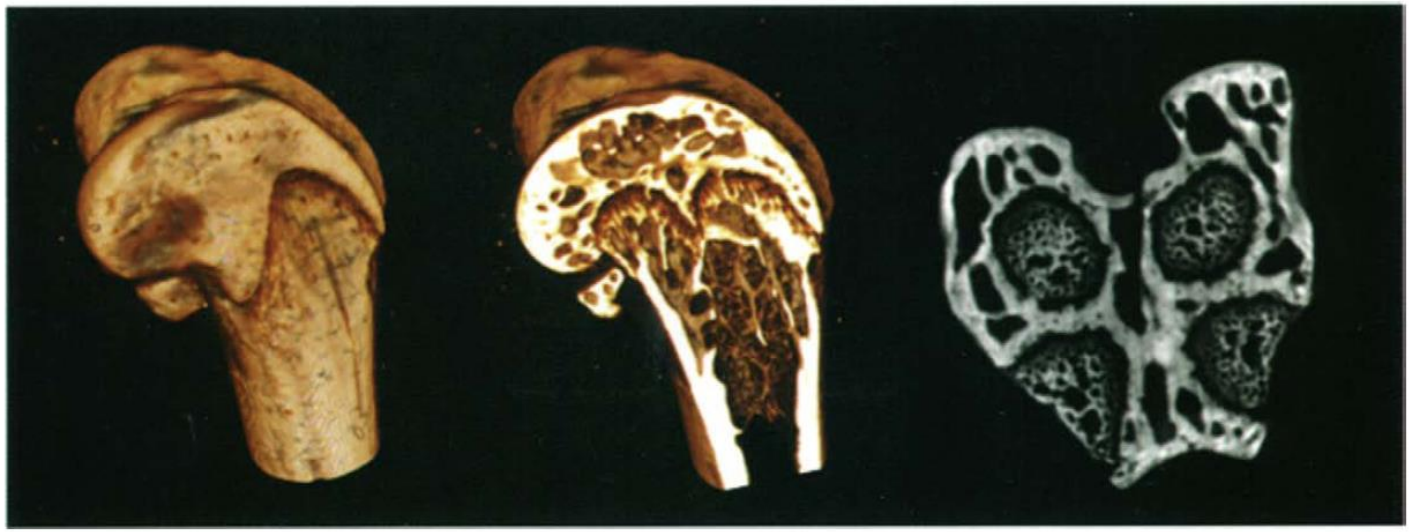


Fig. 14-5 High-resolution microCT imaging of the head of a mouse femur, showing volume rendered images from the 3D dataset (left and center), and a single 2D slice through the femur (right). (Images courtesy of Siemens Preclinical Solutions, Knoxville, TN.)

Ultrasonography

- Use the echo of high-frequency sound to create visual images of internal structures
  - Discriminate cystic vs. solid masses
  - Interrogate body cavities filled w/ fluids (prohibit radiographic techniques)
  - Discriminate the texture of suspected solid masses
  - Locate nonradiopaque calculi w/i organs
  - Biopsy guidance for internal masses & cystocentesis
  - Evaluate cardiac (echocardiograph) [myocardial infarction in mouse], muscle, tendon
  - Diagnose pregnancy (ferret, NHP)

Ultrasonic Imaging

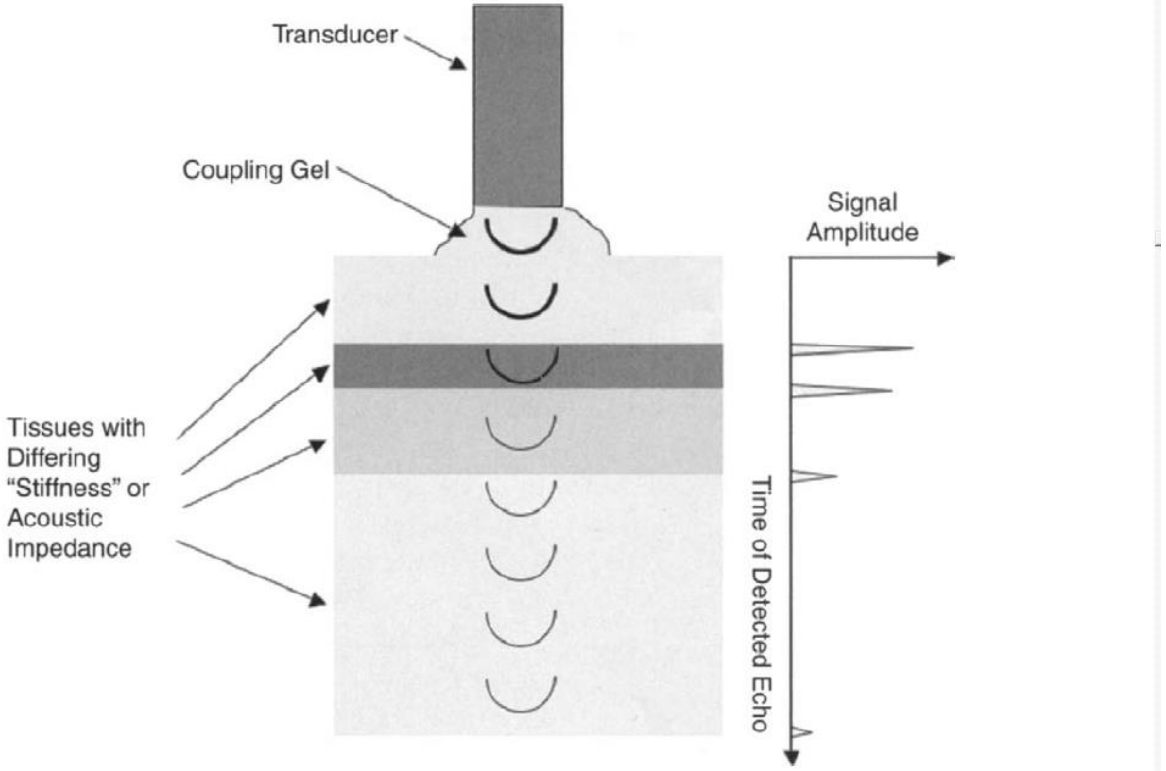


Fig. 14-6 Basic properties of ultrasonic imaging. Ultrasound is emitted by the transducer and part of the energy is reflected back to the transdu aries inside tissues with differing acoustic properties. The time at which this "echo" is received by the transducer provides information on the boundary between the tissues. The ultrasound beam is scanned across the tissue to form 2D or 3D images.

Real-time imaging (ultrasound-guide injection)

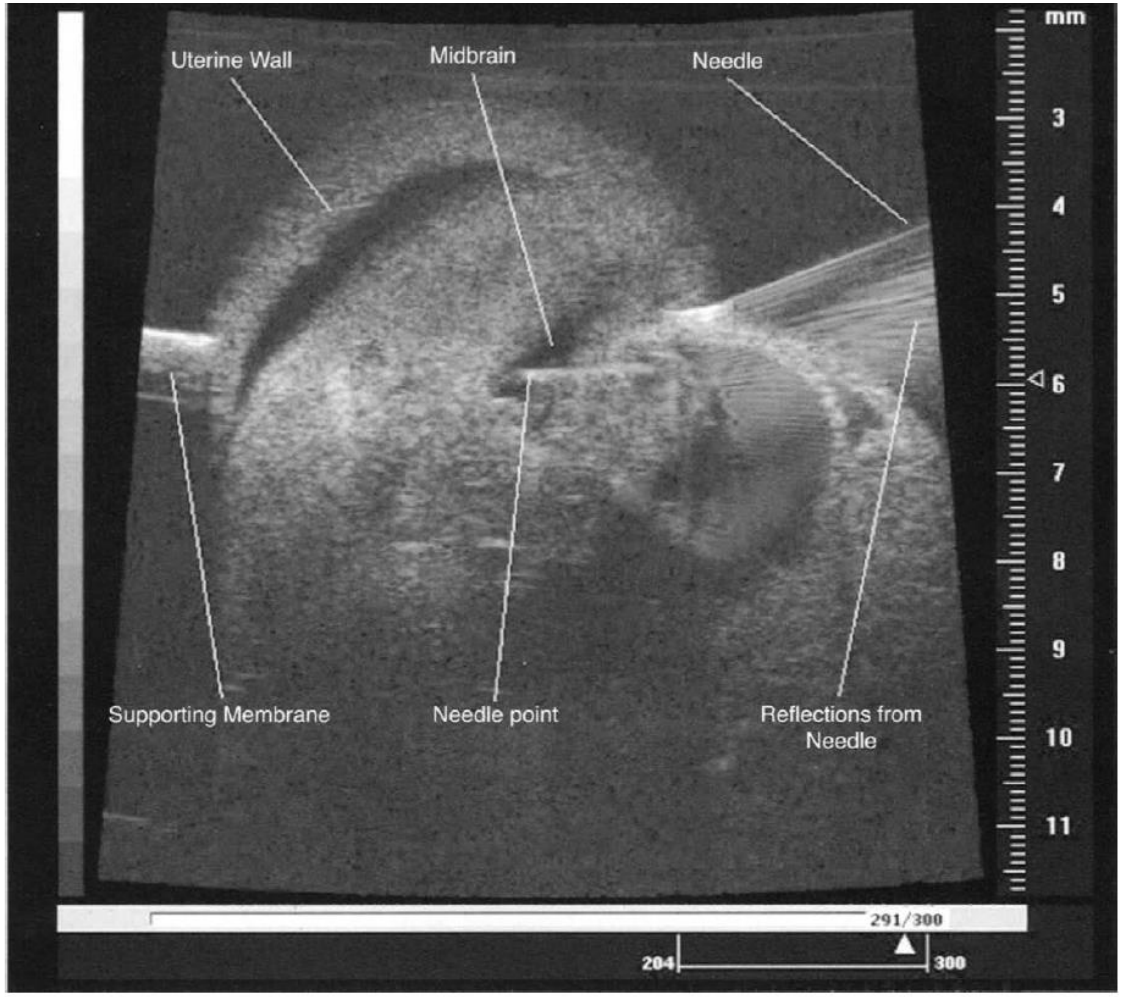


Fig. 14-8 Real-time imaging showing ultrasound-guided injection into the midbrain of a 11.5-day-old mouse embryo. (Image courtesy of VisualSonics Inc., Toronto, ON, Canada.)

MRI (Magnetic Resonance Imaging)

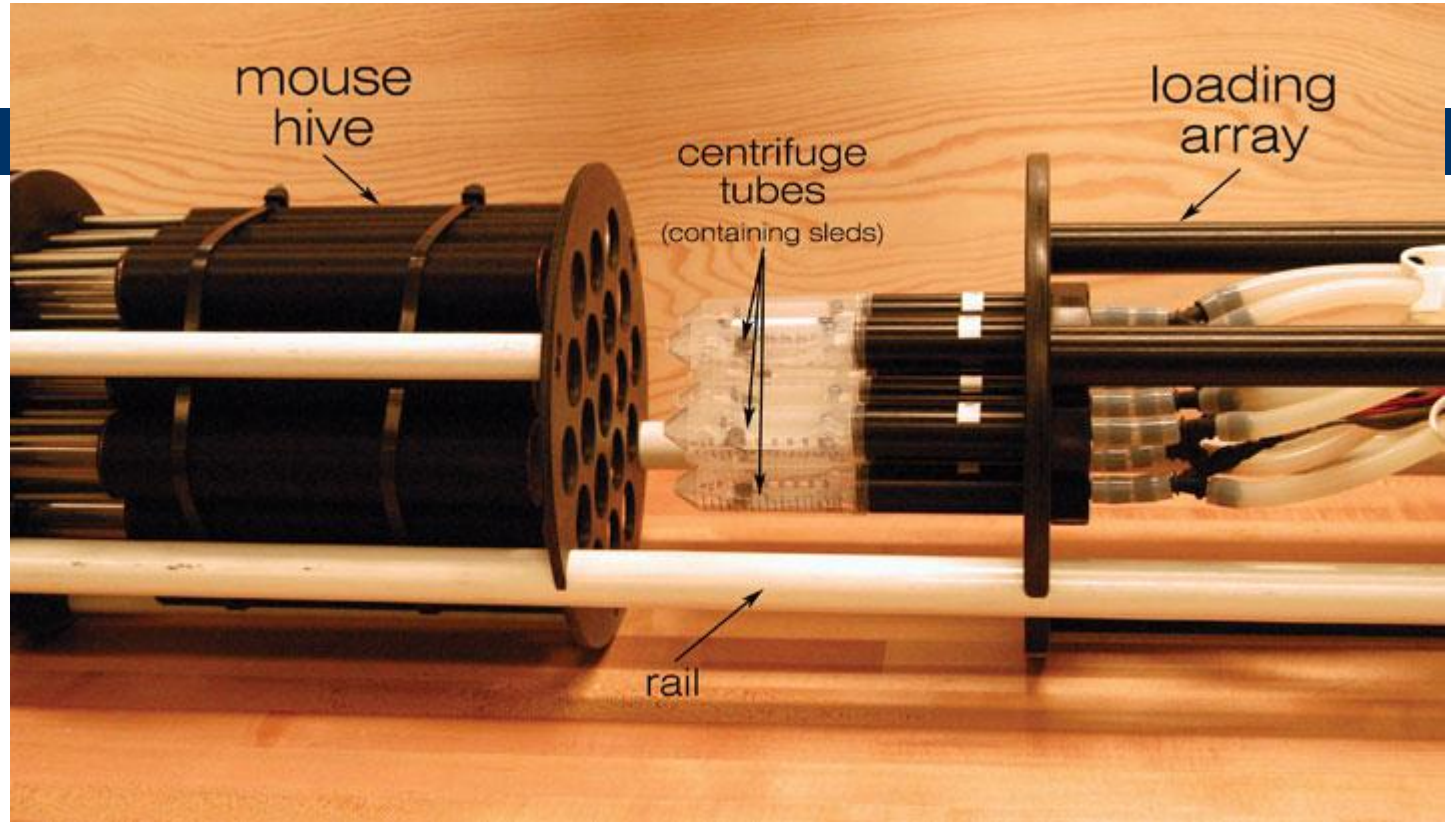
- Previously called nuclear magnetic resonance (NMR) imaging
- In a magnetic field, tissues can be excited when irradiated w/ radiowaves at the resonant frequency determined by the magnetic field strength, then the excited tissue emits a radio signal that can be detected by a coil placed about the tissue, then be used to create an image.

MRI

- A powerful & flexible in-vivo imaging tool
- Freq. to evaluate the head, pelvic structures, bone/joint evaluations
- Commonly for larger animals, rabbits, cats, dogs, NHP
- MRI images of entire mouse and rat bodies using small-bore & strong-magnet machine [4.7T, 7T, 9.4T, 11.6T] (Mouse Imaging Facility)



7T Mouse MRI


mouse
hive
centrifuge
tubes
(containing sleds)
loading
array
rail

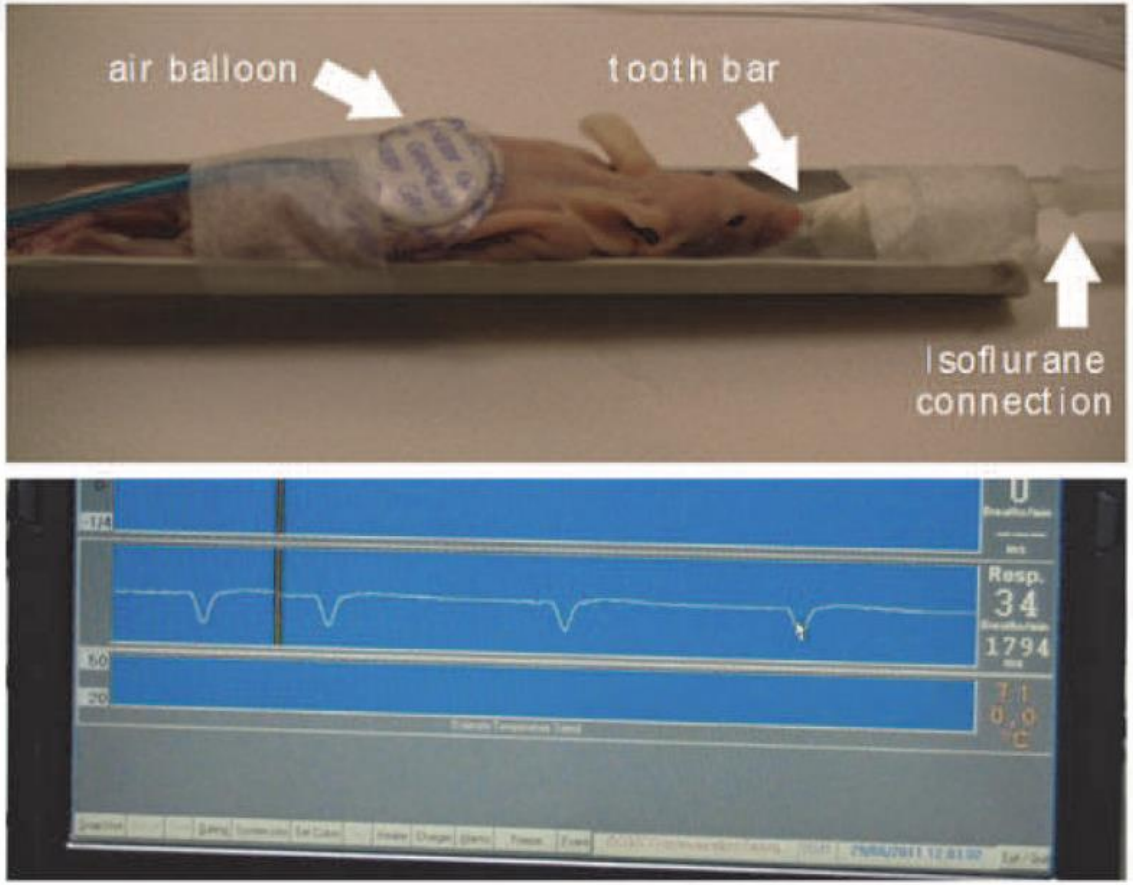
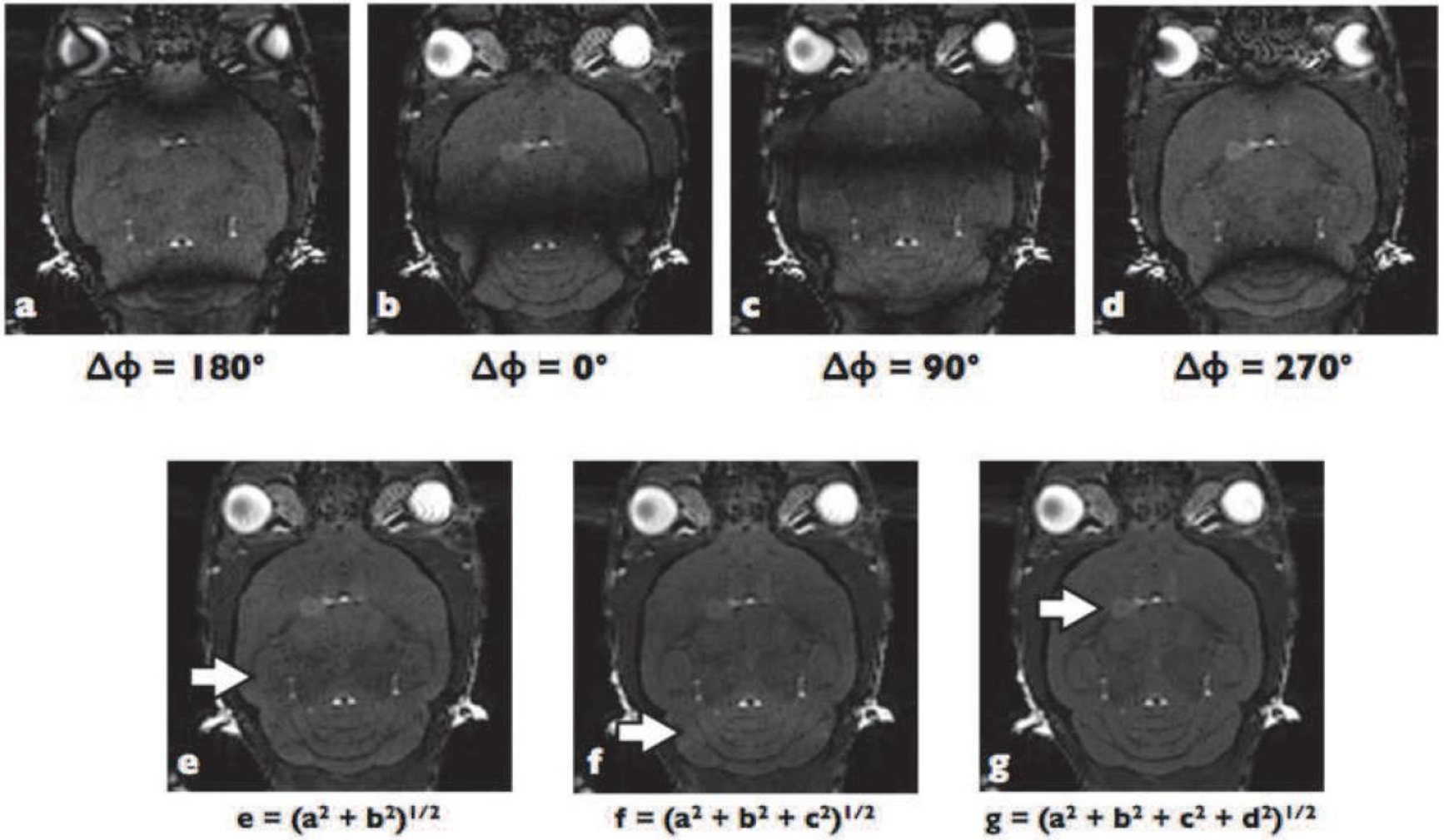


Fig. 3. Nude mouse in a home-made bed. A tooth-bar assembly holds the animal in position and a 1-2% air/isoflurane mixture is delivered in the region of the nose. A balloon is placed on the animal's back to measure its respiration rate. The SA Instruments system is linked to a computer, where the animal's respiration can be visualized on-screen.

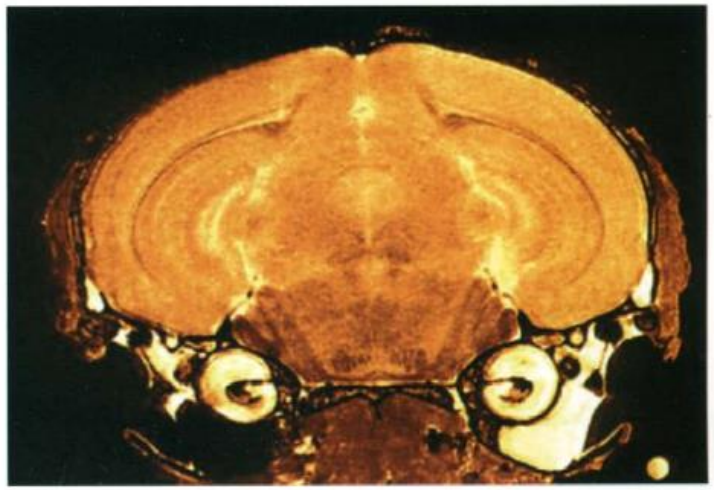
- **Anethetized w/ isoflurane**
- **Resp rate: 70-110 breaths/min**
- **BT (blanket/hot water) @ 32 ℃**



4.7T MRI (mouse brain)



A



B

Fig. 14-10 (a) 3D whole-body T1-weighted MR image of a mouse perfused with a Gd-based contrast agent. (b) High-resolution MR image of a single slice through the mouse brain. (Images courtesy of Prof. Mark Henkelman, Mouse Imaging Center, Toronto, ON, Canada.)

Nuclear Imaging

- functional or molecular imaging approach
- radiolabeled molecules or radiotracers [to specific targets or pathway] to provide contrast
- no background in body

1. Single photon imaging: utilize radionuclides that decay by emitting one or multiple gamma-rays or x-rays; **if combined w/ computed tomographic (rotating gamma camera) →SPECT**
2. PET: utilize radionuclides that decay by **positron emission (two photons w/ strict angular correlation; 180˚ apart)**

Nuclear Imaging (SPECT & PET images)

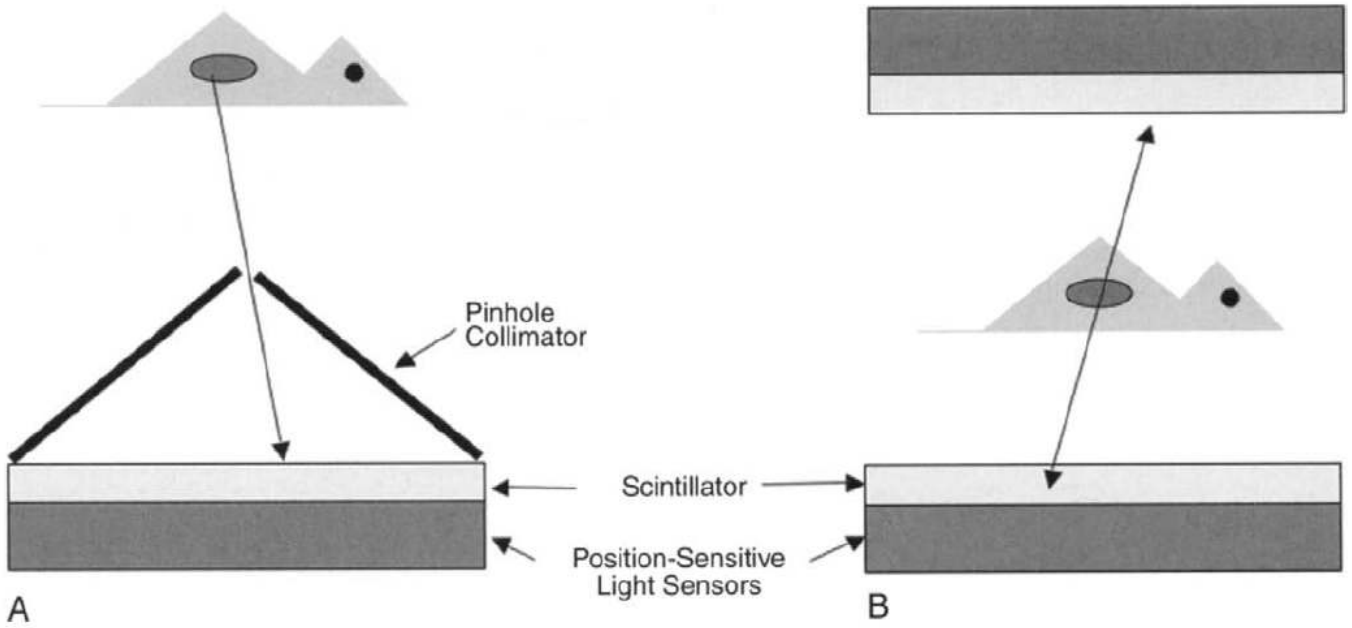


Fig. 14-12 Basic principles of event localization in nuclear imaging. (a) To image radiopharmaceuticals that emit single gamma rays, a collimator is used to define the direction from which detected gamma rays originate. In this example, a pinhole collimator is shown, and all gamma-rays that do not pass through the hole of the collimator are absorbed and do not reach the detector. To obtain tomographic SPECT images, the detector system is rotated around the mouse. (b) Radiopharmaceuticals that decay by positron emission subsequently produce two 511-keV photons that are emitted almost exactly 180 degrees apart. Coincidence detection of these two photons defines the direction of the emitted photons without the need for collimation. Stationary, annular rings of detectors that surround the mouse are typically employed in a PET scanner.

SPECT images

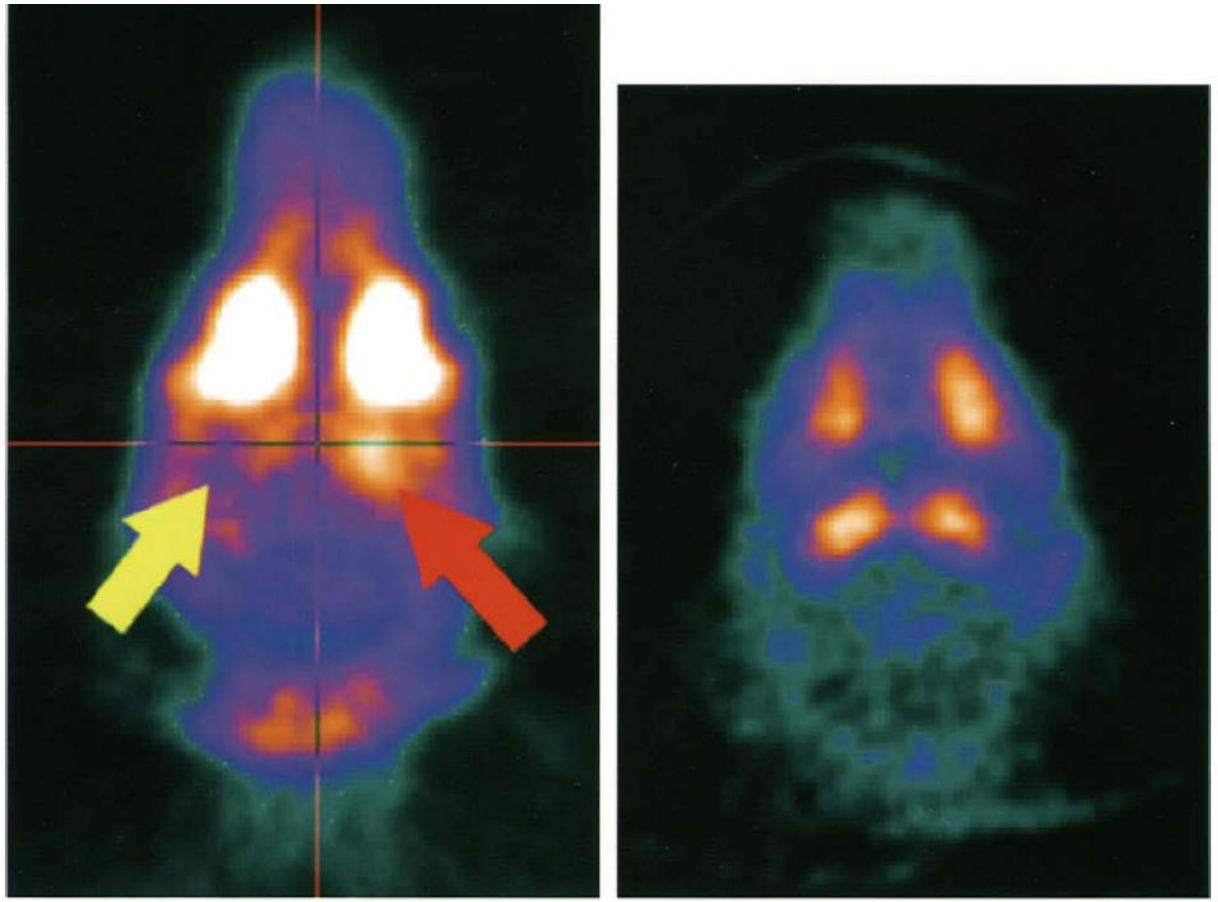


Fig. 14-13 SPECT images of the dopaminergic system in a hemi-lesioned mouse model of Parkinson's disease. (a) Image of dopamine transporters (DAT) acquired using the radiotracer ^{99m}Tc-TRODAT-1. The lesioned side of the striatum (yellow arrow) shows almost total destruction of available DAT, while the unlesioned side (red arrow) shows normal tracer uptake. (b) Image of dopamine D2 receptors acquired using the radiotracer ^{123}I-IBF. Both lesioned and unlesioned sides show normal tracer uptake, indicating that lesioning has only affected the presynaptic neurons, leaving the post-synaptic neurons intact. (Images courtesy of Dr. Paul Acton, Thomas Jefferson University, PA.)

PET images

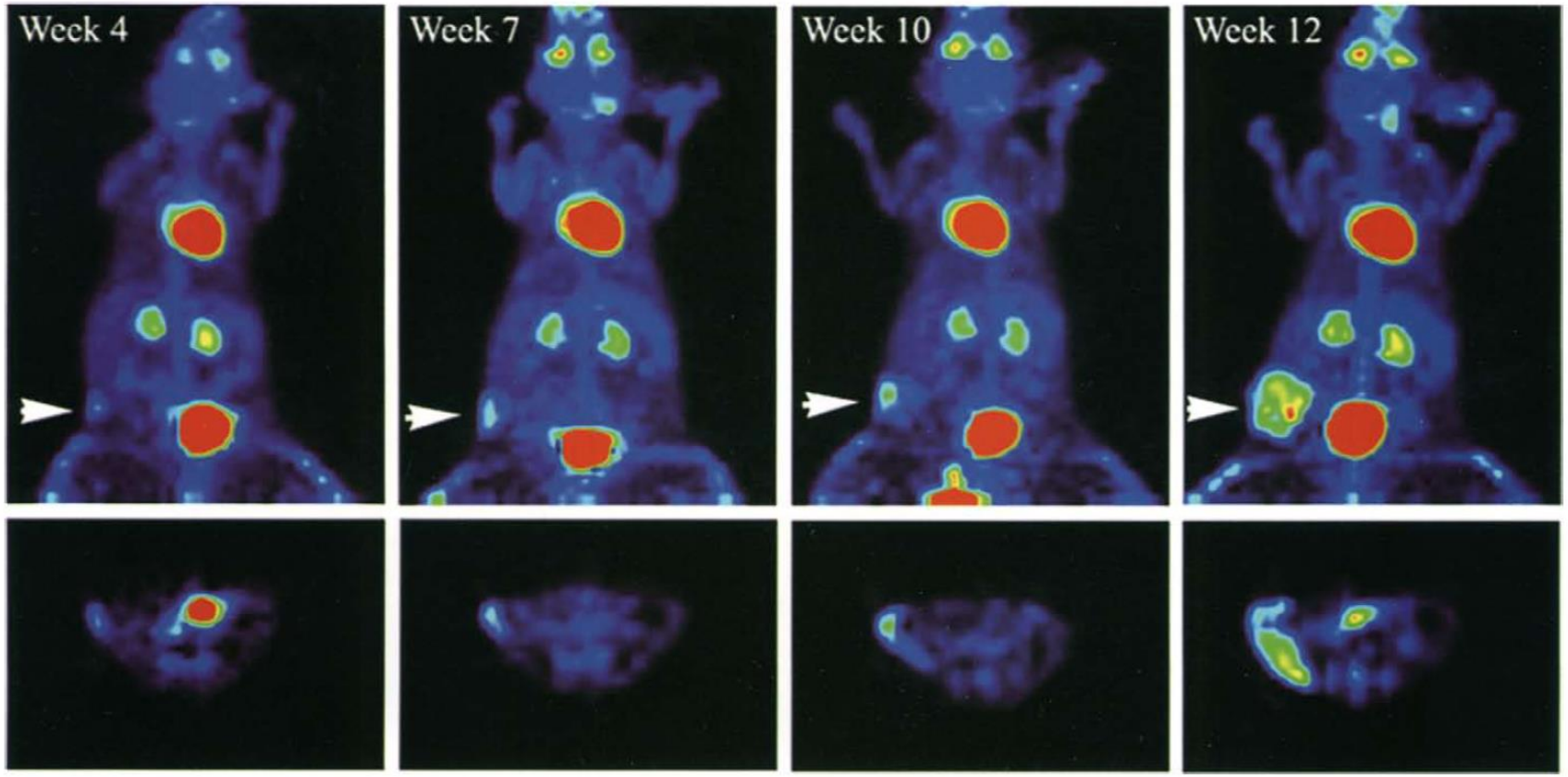


Fig. 14-14 PET images showing tumor initiation and progression in a mouse model of breast cancer. Images acquired using the radiotracer ^{18}F-FDG, a glucose analog. The accumulation of this radiotracer is reflective of local glucose metabolism. (Adapted from Abbey CK et al 2004.)

Optical Imaging

- Mostly widely used radiation for in-vivo whole-body imaging of the mouse is

visible light

- **Two major branches: Fluorescence imaging and bioluminescence image**
- Fluo: injected w/ fluorescently tagged contrast agent or carrying tissues expressing fluorescent transgene
- Biolumin: mice w/ bioluminescent reporter gene, then injected substrate for bioluminescence reaction into mice
- Image in light-tight box [light source & filter for Fluo system]

Fluorescence & Bioluminescence

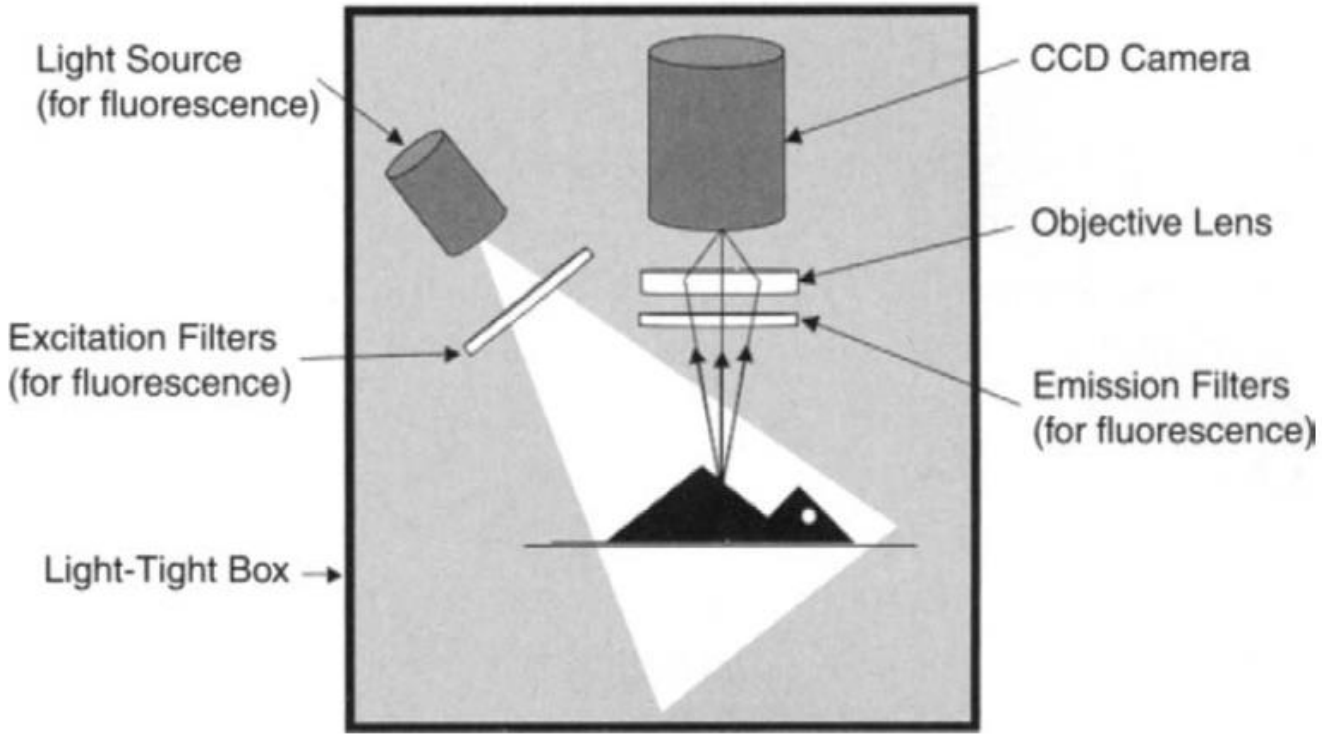


Fig. 14-15 Schematic diagram of optical imaging system for whole-body mouse fluorescence and bioluminescence imaging.

Fluorescence Imaging

- usually utilize Fluorescent reporter gene
- Best known: Green fluorescent protein (GFP)
- Best suitable for whole-body in-vivo imaging is **red part of the spectrum**!
  - DsRed
  - Monomeric red fluorescent protein (mRFP1)
- possible only use fluorescently taged contrast agents w/o reporter gene in cells/tissues

Fluorescence imaging

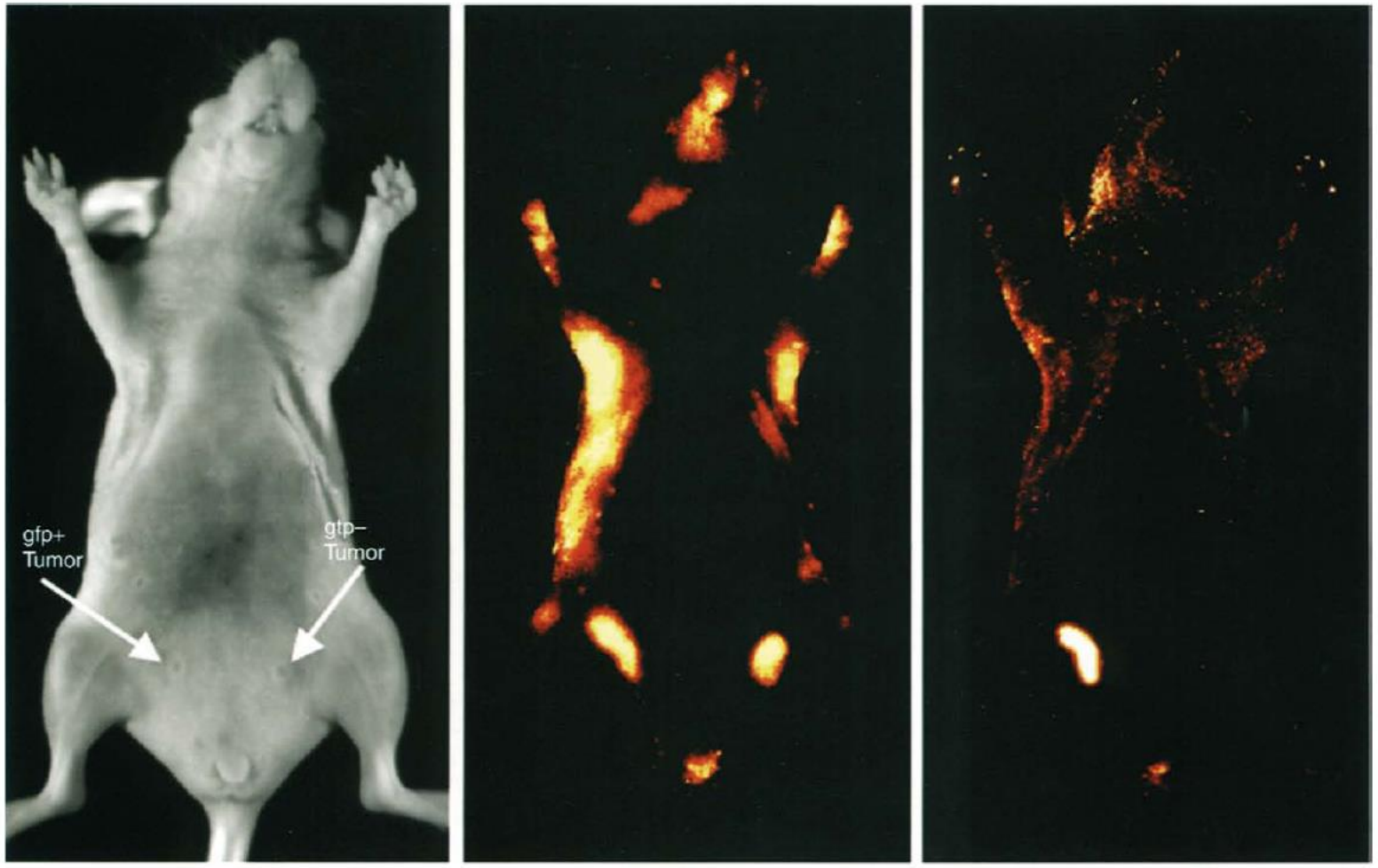


Fig. 14-17 Fluorescence imaging of 9L glioma tumors implanted in the flank of a mouse. The mouse was injected with a Cy5.5-based fluorescence probe that is activated only by proteases such as Cathepsin B which are found at high concentrations in tumors. The tumor on the left also carries a green fluorescent protein (gfp) reporter gene. Two separate acquisitions with appropriate excitation and emission filters were used to obtain separate images of the Cy5.5 probe and gfp. (Image reprinted with permission from Mahmood, U., Tung, C-H., Tang, Y., Weissleder, R. (2002) Feasibility of in vivo multichannel optical imaging of gene expression: experimental study in mice. *Radiology* **224,** 446–451.)

Bioluminescence Imaging

- Need a gene encoding an enzyme to catalyze a bioluminescent reaction in animal
- **very weak** signal (but no background in mice)
- Two systems: firefly luciferase (630 nm, 37C) and renilla luciferase (482 nm)
- Applied in
  - transfected cells for tumor study
  - into infectious agents (virus/bact.) for infectious D study
  - gene expression study

Bioluminescence imaging

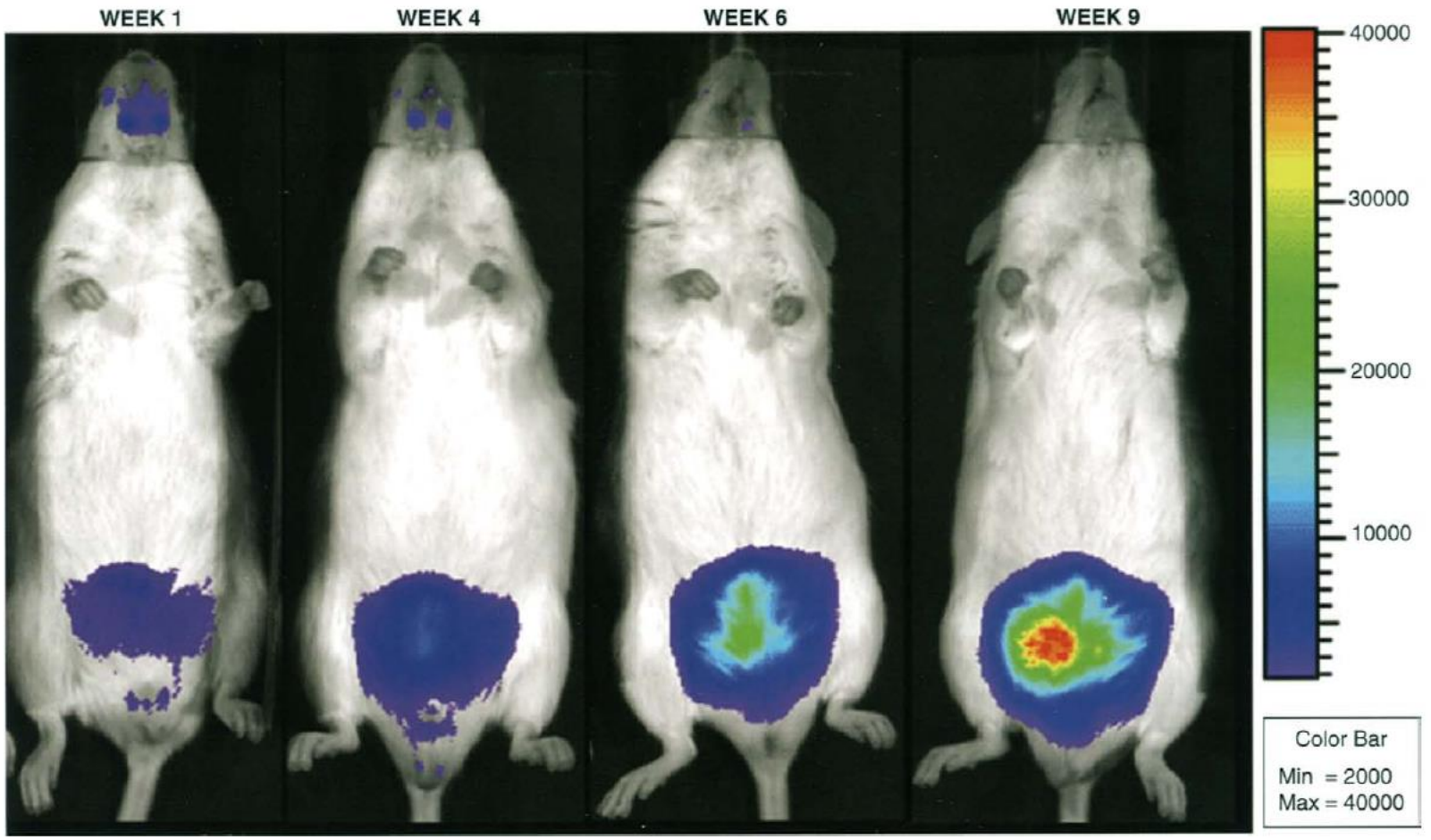
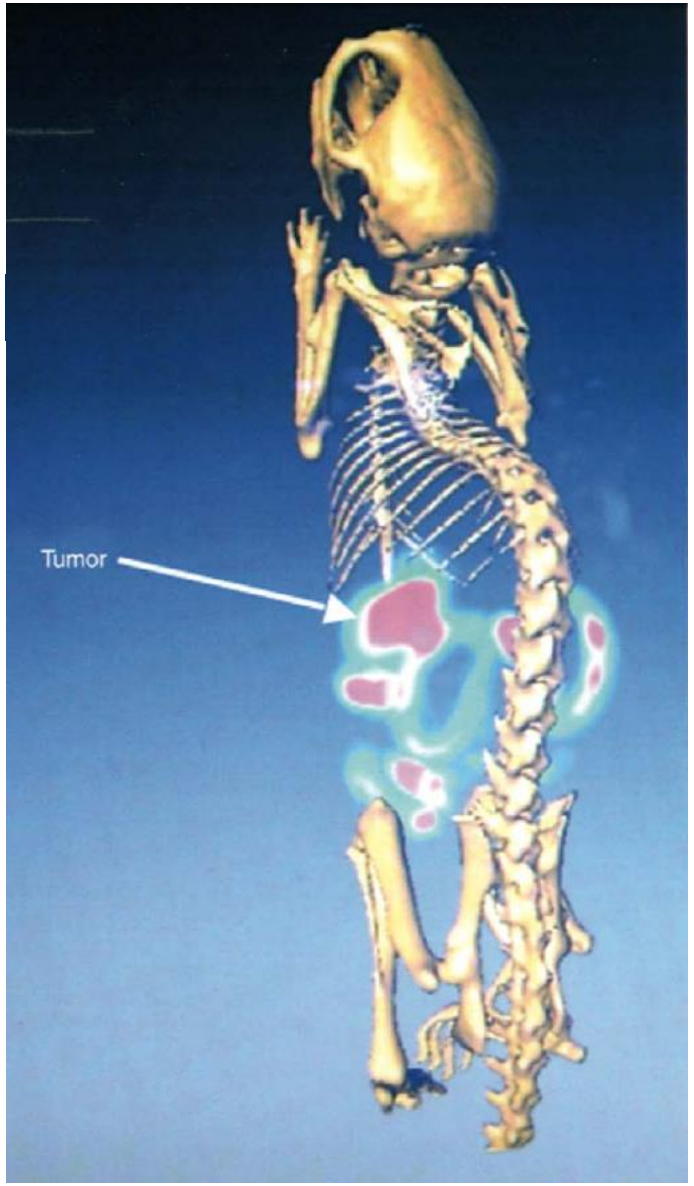


Fig. 14-16 A series of bioluminescent images of the same individual adult mouse taken over time. This transgenic mouse expresses firefly luciferase in the prostate gland and is developing a SV40 T and t antigen induced prostate tumor. Light emission is proportional to the number of luciferase expressing cells in this mouse, thus a relative increase in brightness over time is indicative of prostate tumor development. The intensity of light emission in each image is depicted as false colors that correspond to the scale bar (showing relative light units) on the right side of the figure. (Images courtesy of Dr. Scott Lyons, Xenogen Corporation, Alameda, CA.)

Multimodality Imaging

- Utilize multiple imaging modalities to study animal models
- Get [SPECT/CT] [PET/CT] [PET/MRI] data and 2D imaging system (images from fluorescence, bioluminineSCence, x-ray signals
- Ex. targeted fluorescent probe/radiotracer to co-register microCT or MRI image
- Combined SPECT/CT data in 3D to show bone (CT) and biodistribution of interested labeled material/tissue (ex. radio-labeled)

➢Combined SPECT/CT data in 3D to show bone (CT) and biodistribution of a ^{99m}Tc-Labeled tumor-seeking radiopharmaceutical.



Whole-Body In-Vivo Mouse Imaging Modalities

TABLE 14-1

SUMMARY* OF MAJOR WHOLE-BODY IN-VIVO MOUSE IMAGING MODALITIES

Modality Resolution	Spatial Sensitivity	Contrast Agent	Time to Acquire Image	Radiation Dose	Cost
X-ray radiography	~20–100 μm	~mM	~sec	~1–50 μGy	$
X-ray CT	~20–250 μm	~mM	~min	~1–20 cGy	$$
Ultrasound	~50–200 μm	?	real-time	n/a	$–$$
MRI	~50–500 μm	~mM–μM	~min–h	n/a	$$–$$$
SPECT	~0.2–4 mm	~nM–pM	~min–h	~1–100 cGy	$$
PET	~0.8–2.5 mm	~nM–pM	~min–h	~1–100 cGy	$$–$$$
Bioluminescence	~mm	~nM	~sec–min	n/a	$–$$
Fluorescence	~mm	~nM	~sec–min	n/a	$$

CT, computed tomography; MRI, magnetic resonance imaging; SPECT, single photon emission computed tomography; PET, positron emission tomogra

Consideration for Lab Animal Imaging Center Design & Setup

- Facility design (general consideration, animal housing & supply, MRI-specific consideration..etc.
- Animal imaging support
- Personnel
- Imaging equipment
- Data management

Facility Design- general facility considerations

- Traffic patterns (circulation)
  - minimized cross-circulation between animals and personnels
  - health status of lab animals (subclinical infections)
- Supply storage rooms (usu. under-estimated!)
- Administrative & Personnel offices
- Break rooms
- Restrooms and showers (biosecurity level of animal facility)
- Conference room
- Environmental considerations (animal/personnel area w/ higher humidity and running equipment will generate lots of heat

Animal Housing & Imaging Support

- Ideal: imaging center incorporates animal housing space or adjacent to animal holding facility
- Animal Housing: size & type of housing room (depend on species)
  - rodents, NHP, …
  - space for temporary quarantine of animals (before/after imaging)
- Procedure Rooms: usu. 1:4 ratio of procedure rooms vs animal holding rooms
- Cagewash: “dirty” vs “clean” sides
- Husbandry supply storage
- Euthanasia

Magnet-Specific Facility Design

- Location
  - magnetic forces can inactivate/alter life-supportive devices (pacemaker, insulin pump)! (health hazard)
  - magnetic forces may interfere other equipment
- Structural considerations
  - MRI (radiofrequency **RF** to generate images) might be interfered by the RF of other devices. So, **in RF-shielded rooms**!
- Cryogen gases: need liquid Helium and liquid Nitrogen
- Environmental Consideration: stable temp; continual monitoring temp, humidity, and oxygen levels
- Work area and tools

Miscellaneous Considerations

- Radiation Safety
- Animal Biosafety
- Laminar flow hoods
- Ducted biosafety cabinets
- Sterilization

Animal Imaging Support

- Anesthesia Equipment
- Physiological Monitoring
- Surgical Support

Personnel

- Imaging specialist
- Animal Support
- Computer Information Technology
- Housekeeping & Facility Maintenance
- Unique MRI Personnel Safety Consideration

Imaging Equipment

- MRI
- X-ray Computed Tomograph (CT)
- Positron Emission Tomograph (PET)
- Ultrasound Imaging
- Optical Imaging

Data Management

- Data Processing
- Data Analysis
- Data Storage



http://www.mic.ntu.edu.tw/tw/about/organization.html

Reference

- The Mouse in Biomedical Research (e-book in NTU Library)
  http://tulips.ntu.edu.tw/search~S5*cht?/XMouse+in+biomedical+research&searchscope=5&SORT=DZ/XMouse+in+biomedical+research&searchscope=5&SORT=DZ&SUBKEY=Mouse+in+biomedical+research/1%2C58%2C58%2CB/frameset&FF=XMouse+in+biomedical+research&searchscope=5&SORT=DZ&2%2C2%2C
- Considerations for lab animal Imaging Center Design and Setup
  http://ilarjournal.oxfordjournals.org/content/49/1/4.full.pdf+html

Thank You for Your Attention!

萬灼華副教授
國立臺灣大學獸醫專業學院
chwan@ntu.edu.tw







